

Determinism and Free Will

Peter Trinh

CS 6795 Intro to Cognitive Science

Georgia Institute of Technology

ptrinh32@gatech.edu

I. DETERMINISM AND FREE WILL

For this prompt I will be using ChatGPT as my chosen LLM. This large language model generate text based on predictions made from the likely next token from context, which the temperature controls the variability [1]. Following the working definitions given for determinism and free will from the prompt, I tested this by running the prompted example "Describe a sunrise in two sentences" within ChatGPT at temperature levels of 0.0, 0.4, and 0.8. At 0.0 the model always pick the most probable next token, which means that if the prompt would be repeated multiple times, then the results will be exactly the same wording and phrases every single time. This is known to be the most deterministic. The results from the test of temperature 0.0 were:

"The sky gradually brightens with shades of pink and orange as the sun rises above the horizon. The world awakens to a warm, golden light."

At temperature of 0.4, there would be slight randomness from the results of the prompt. There would still be a decent flow and natural response from this level of variability. However, multiple runs of this level would result in similar answers from ChatGPT and no identical responses. The results from the test of temperature 0.4 were:

"The sky glows with soft hues of pink and orange as the sun slowly appears above the horizon. Morning light spreads warmth across the quiet landscape."

With a temperature of 0.8, the token would be greatly randomized and the choices would be strongly different. The results would be a great varied answer that fluctuates in text, tone, and overall sentence structure with every run. This is the results from the test of temperature 0.8:

"Amber and rose streaks scatter across the morning sky as the sun ascends. Birds stir, and the air shimmers with the promise of a new day."

The experiment clearly shows that the model's behavior is not truly free. Even though at level 0.4 and 0.8 there is variable within the responses, the randomness that contributes to the answers does not mean there is free will in the system. The deterministic rules that can be seen in temperature 0.0 are still affecting the LLM's system and can be seen more as controlled randomness instead of genuine free will. There is no possibility of free will within the system.

II. AUTONOMY

For this prompt I continued using ChatGPT as the LLM. The first prompt I gave was "Write me a generic how are you doing email to send to a coworker". The second prompt given changed the condition to "Now write one I can send to my best friend". Both responses can be seen within the appendix.

Based on the responses from ChatGPT, can be seen that the LLM was able to show some adaptability within it's answer. When the conditions from "coworker" was changed to "best friend", the LLM was able to adjust the overall tone of the generated email from a more professional stance to a more casual answer. This shows that it was able to recognize and adapt to the newly given situation based on the context that was given by the user.

Although the LLM was able to adjust accordingly, the model itself does not seem to be autonomous. The language model does not truly understand the meaning behind the context. It generates the text and adjusted the tone by recognizing previous statistical patterns instead of autonomous thought [2]. Ultimately, the LLM is capable of reacting to certain situations based on given context and adapting to said situation accordingly from drawn statistical inferences; however, this does not equate to true autonomy. The model isn't capable of decisions for itself and won't have the capacity for such actions.

III. SENTIENCE

Free will and autonomy are not necessarily requirements of for sentience. Sentience is about having subjective experiences, based on the prompt. The difference that separates this distinction would being able to act on them or not. For example, there are many cases in which paraplegic patients are known to still be able to feel and experience the world around them; however, they simply are not able to respond to such feelings [3]. Having sentience does not equate to having free will or autonomy and vice versa. Free will and autonomy are more aligned to be results from sentience or effects drawn from sentience instead of requirements.

Based on conclusions and research from the first two prompts, LLMs lack the ability for sentience overall. For example, ChatGPT does not have the ability to choose or have intention behind it's actions. It's actions are restricted by a governing set of rules, leaving no ability for free will. It has no self intention or autonomy outside its prompts. Coupled with the inability for subjective experience, LLMs like ChatGPT do not have the capacity for sentience.

Considering the experiment, if we were to design a system capable of detecting sentience, I believe the system itself must be capable of sentience. The ability for sentience is the defining factor in which sentience can be confirmed. We as people recognize sentience in other people and animals, but this is based on external factors that is observed and analyzed. Things like thought processes, feelings, and pain are all indicators to us that a being is sentient in itself. However, this is still an educated guess based on these observations. The best way to reliably detect sentience would be to share experiences and confirm each other's experiences. Subjective experiences may be unique to each person, but the ability to feel the experience itself is how sentience can be confirmed. This means that for a system to be sure of sentience, it must have sentience itself.

REFERENCES

- [1] Buck Shlegeris, Fabien Roger, Lawrence Chan, and Euan McLean, "Language Models Are Better Than Humans at Next-token Prediction," *ArXiv Preprint arXiv:2212.11281*, Dec. 2022
- [2] E. M. Bender and A. Koller, "Climbing towards NLU: On meaning, form, and understanding in the age of data," *Proc. 58th Annu. Meeting Assoc. Comput. Linguistics*, 2020.
- [3] J. Cole, "The phenomenology of agency and intention in the face of paralysis and insentience," *Phenomenology and the Cognitive Sciences*, vol. 6, no. 3, pp. 309–325, 2007.

APPENDIX A CHATGPT AUTONOMY RESPONSE 1

Subject: Checking In

Hi [Name],

I hope you're doing well! It's been a little while since we last caught up, so I just wanted to check in and see how things are going on your end. How have you been?

Things have been pretty busy on my side with [optional: brief mention of your current project or task], but overall going well. Would love to hear how everything's been for you — both work and otherwise.

Take care,

[Your Name]

APPENDIX B CHATGPT AUTONOMY RESPONSE 2

Subject: Just checking in

Hey [Name],

It's been a minute! I was just thinking about you and wanted to check in to see how you've been doing lately. How's life treating you? Anything new or exciting going on?

Things on my end have been [optional: short personal update — e.g., "pretty busy but good," "a bit of a rollercoaster lately," etc.]. I miss catching up like we used to — we should definitely find a time to hang out or call soon.

Hope everything's going great with you. Seriously, tell me what's new — I want all the updates!

Talk soon,

[Your Name]